Assignment 2

1. Self-rating in LLM, Deep Learning, AI and ML

LLM(B) :- I understand the basics of large language models and have worked with Gemini APIs and LangChain in projects and have built end-to-end applications using them.

Deep Learning(B) :- I have worked with TensorFlow, Keras, YOLOv8 and EfficientNet-B0 for image classification and object detection.

AI(B) :- I know the fundamentals of AI and have applied them in projects involving computer vision and generative AI.

Machine Learning(B): - I have knowledge ML concepts, data preprocessing, model training and evaluation.

2. What are the key architectural components of an LLM-based chatbot?

LLM-based chatbot has a few main components:- The user sends a question through the interface, backend receives it and sends it to the LLM which understands the question and generates a response. If the chatbot needs information from documents or other data we can use a retrieval system to find relevant information and provide it to the LLM. Conversation history can also be stored to maintain context. The response is sent back to the user.The main components are the UI, backend, LLM, prompt management,retrieval system, conversation memory and database. The exact architecture depends on what the chatbot is designed for.

3. What are vector databases? Which one would you choose and why?

Vector database stores information as numerical values called vectors or embedding. They capture the meaning of text, images and other data. if a user searches something the database finds the information using a similarity search instead of matching only the exact words,if I had build a chatbot that answers questions from something simple like a notes, I would choose ChromaDB since it is open source and easier to set up. I will convert the notes into embeddings, store them in ChromaDB and retrieve the most relevant portion of it when a when a question is asked and then given to the LLM to generate an answer based on it.For a bigger application, I would also compare options based on scalability and performance